GAME THEORY

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This is to certify that that the project work entitled "GAME THEORY" submitted by Nandana M Nair to CMS College Kottayam in the partial fulfilment of the requirements for the degree of Bachelor of Science in Mathematics, is a bonafide work carried out under my supervision and guidance.

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CERTIFICATE

This is to certify that Nandana M Nair has undergone the B. Sc. Degree Course at CMS College Kottayam during the period 2021-24 and have undertaken the project under Mrs. Sindhu Thomas, Department of Mathematics, CMS College Kottayam.

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INTRODUCTION

Game theory involves making decisions based on the choices, actions, and potential payoffs of all parties involved, with the assumption that they will act rationally to achieve the best possible outcomes. The theory provides a framework for analyzing and understanding various strategic situations.

Game theory had its origin in 1928 by Jon Von Neumann, who wrote the book *Theory of Games and Economic behaviour* along with the economist Oskar Morgenstern. The publication of this book is seen as the initial point of modern game theory.

Game theory, a branch of mathematics, examines strategic interactions between rational decision-makers in competitive situations, providing a mathematical analysis of conflicting interests to determine optimal choices for achieving desired outcomes under specified conditions. Game theory delivers an outline based on the construction of difficult models that describe the situation of conflicts and cooperation between rational decision makers. Game theory is connected with human interaction and at the same time an objective to maximize profit or to rationalize the decision

Game theory aims to explain the strategic actions of two or more players in a situation governed by a set of rules and outcomes, and can be used to determine the most likely outcomes when the situation involves known payouts or penalties. According to game theory, the actions and choices of all participants influence the outcome, and it is assumed that players are rational and will strive to maximize their payoffs.

Game theory is utilized in various sciences, including biology and sociology, to analyze long-term situations. Game theory has been primarily used to address economic issues and has been successfully applied in various contexts, including business competitions, market dynamics, political campaigns, jury voting, auctions and procurement contracts. It is also applied in psychology and evolutionary biology. Game theory also has relevance in economics, political science, and international politics in particular.

CHAPTER 1

WHAT IS A GAME?

A game is a formal explanation of a tactical situation. Activity between two or more gain for each other or suffers loss. The word "game" describes any situation involving positive or negative outcomes determined by the player's choices. A game is a depiction of strategic interaction that encompasses the limitations on the actions available to participants and their interests, but does not delineate the precautions that players may adopt.

1.1. ELEMENTS OF GAME

- Players: They are the set of participants who make the decisions.
- Rules: Set of instructions based on which the players make decisions.
- Strategy: Plan of actions taken by a player based on the rules provided to gain maximum outcome.
- Payoffs: they are the scores earned by each player at the end of the game.
- Outcomes: consequences of each possible set of actions by players.

The game identifies the players identities, preferences, available strategies and how these strategies affect the outcome.

1.2. FEATURES OF GAME

- Competitiveness:

A situation is said to be competitive if:

- The number of competitors is limited
- All possible outcomes are known for each player in the game.
- No player can know other player's decision before making their decisions.
- The gain made by each player may be positive negative or zero. A negative gain is called loss.
- Number of people
- There is a finite number of players or competitors.
- When the number of persons playing is 'n' then the game is known as 'n' person game.
- The players simultaneously select their course of action.
- The number of moves done by each player may finite or infinite. Accordingly, each player may have finite or infinite choices.
- Players have access to information regarding the previous actions of other participants. In this case either complete or partial or no information may be available.

1.3. ASSUMPTIONS OF GAME

- The players act rationally i.e. the players are behaving consistently with their preferences.
- Each player knows everything about the game.
- A finite art of possible course of action is available with each player.
- The players aim to maximize gain and minimize losses.
- The payoff is pre-determined and fixed.
- A player can adopt multiple strategies in the game.
- The payoffs must represent utilities.

Fair Game

A game is a fair game if the value of the game is zero, i.e. there is no loss or gain for any player in the game.

1.4. BASIC CONCEPTS OF GAME THEORY

1.4.1. STRATEGY: Strategies refers to the plan of action done and executed by each player to earn maximum gain from the game. There are mainly three types of strategies:

- Pure strategy
- Mixed strategy
- Optimum strategy

1.4.2. PAYOFF MATRIX: A payoff matrix is a matrix that indicates the resulting gain in two person game, it represents the outcomes and the payoffs of different strategies acquired by the participants.

Structure of Payoff Matrix:

- Rows: Representing the strategies of player 1
- Columns: Representing the strategies of player 2
- Cells: Containing the payoffs for each combination of strategies

E.g.:

PLAYER Y

PLAYER X

	Y1	Y2	Y3
X1	24	36	8
X2	3	20	16

1.4.3. SADDLE POINT: A saddle point describes the solution of the game.

An element in a payoff matrix that is the minimum in its row and the largest in its column is called a saddle point.

Saddle point: minimax = maximin

MINIMAX THEOREM1: Let f (X, Y) be such that both max min f (X, Y) and min max f (X, Y) exist. Then

$$\min \max f(X, Y) \leq \min \max f(X_0, Y_0) \quad (1)$$

proof. Let X_0 and Y_0 be some abitrarily chosen points in $E_n$and E_m respectively. Then

$$\min f(X_0, Y) \leq f(X_0, Y_0)$$

and $$\max f(X, Y_0) \geq f(X_0, Y_0)$$

Hence $$\min f(X_0, Y) \leq \max f(X, Y_0)$$

But Y_0 is arbitrarily chosen and could have been any point in E_m. and for every one of them the inequality should hold. Even if we had chosen Y_0 to be that point for which max f(X,Y) has the least value, the inequality shall be true. So

$$\min f(X_0, Y) \leq \min \max f(X, Y)$$

Also, since X_0 is any point in E_n the inequality will hold even if we choose that X_0 which makes min f(X, Y) maximum. Therefore

$$\min \max f(X, Y) \leq \min \max f(X, Y)$$ Proved.

THEOREM 2: Let f(X, Y) be such that both max min f (X, Y) and min max f (X, Y) exist. Then the necessary and sufficient condition for the existence of a saddle point $f(X_0, Y_0)$ of f (X, Y) is that

$f(X_0, Y_0) = \max \min f(X, Y) = \max \min f(X, Y)$

proof. (I) To prove that the condition is necessary. Let (X_0, Y_0) be a saddle point such that (3) holds. Since

$f(X, Y_0) \leq f(X_0, Y_0)$ for all $X \in E_n$.

$\max f(X.Y_0) \leq f(X_0, Y_0)$

But $\min [\max f(X, Y)] \leq \max f(X.Y_0)$,

Therefore, $\min \max f(X, Y) \leq f(X_0, Y_0)$

Again, from (3) , since

$f(X_0, Y_0) \leq f(X_0, Y)$ for all $Y \in E_m$.,

$f(X_0, Y_0) \leq \min f(X_0.Y)$.

But $\min f(X_0, Y) \leq \max [\min f(X, Y)]$.

Hence $f(X_0, Y_0) \leq \max \min f(X, Y)$.

Thus, we find that

$$\min \max f(X, Y) \leq f(X_0, Y_0) \leq \max \min f(X, Y).$$

But from theorem 1.

$$\min \max f(X, Y) \geq \max \min f(X, Y)$$

The only conclusion from the above two statements is that

$$\max \min f(X, Y) = \min \max f(X, Y) = f(X_0, Y_0)$$

(ii) To prove that the condition is sufficient, let (4) be true. Let the maximum of Min $f(X, Y)$ occurs at X_0, and the minimum of max $f(X, Y)$ occur at Y_0. Then, from (4)

$$\min f(X_0, Y) = \max f(X, Y_0),$$

But by definition of minimum,

$$\min f(X_0,Y) \leq f(X_0,Y_0),$$

and so, from (5),

$$\max f(X, Y_0) \leq f(X_0,Y_0),$$

which means that

$$f(X,Y_0) \leq f(X_0,Y_0) \text{ for all } X.$$

Also, by definition of maximum,

$$\max f(X,Y_0) \geq f(X_0,Y_0),$$

and so, again from (5),

$$\min f(X_0, Y) \geq f(X_0,Y_0),$$

which means

$$f(X_0, Y) \geq f(X_0,Y_0) \text{ for all } Y.$$

Thus, we find that

$$f(X, Y_0) \leq f(X_0, Y_0) \leq f(X_0, Y).$$

which, by definition, means that (X_0, Y_0) is a saddle point of $f(X, Y)$.

Proved.

CHAPTER 2

STRATEGIES IN GAME THEORY

In a game, a strategy is a player's entire strategic plan. It is a set of moves or actions done by the player to gain maximum outcome for himself and to minimize the outcome for other players during the game. A strategy must specify each action a player will take during the game. Each player outcome will be based on these strategies. There are mainly three types of strategies:

- Pure strategy
- Mixed strategy
- Optimal strategy

2.1. PURE STRATEGY

A detailed and complete strategy in which the players intended to follow specific plan of action in every possible situation.

Specificity, completeness and determination are some of the characteristics of pure strategy.

EXAMPLE:

(1) Rock-paper scissors

When two players compete for a single point in the game of rock, paper, scissors, their payoff ma rix can be

	ROCK	PAPER	SCISSORS
ROCK	0	-1	1
PAPER	1	0	-1
SCISSORS	-1	1	0

Suppose both are playing rock paper scissor game, in each round whoever wins will receive one point and whoever lose will receive -1 point, also when the game becomes a tie, then zero will be the receiving point.

While looking the case of A, suppose A is using paper and B is using rock, then A wins since paper can cover the rock, hence A receives 1. Likewise, the game continues receiving -1,1 and also 0 for the player A and B.

2.2 MIXED STRATEGIES

In a strategic game, a mixed strategy occurs when the player makes decisions based on the probability distribution of his options rather than selecting a single, definitive course of action. Each player uses a different strategy at different points during the game.

<u>Problem:</u>

Examine the payout matrix below for Player A and find the best solution.

Player B

Player A

9	7
5	11

Ans :

Player B

Player A

9	7
5	11

Row Min 7 (maxi)

column Max 9(Min)

The Max value ≠ Min value

So, this game has no saddle point

i.e., Maximin ≠ Minimax ; 7≠9

Now we proceed for mixed strategy

oddments

9	7	6
5	11	2
4	4	

I. Oddments - values calculated from payoff matrices to determine optimal mixed strategies for two-player games. Specifically, oddments are the differences between payoff values in opposite cells of a 2x2 payoff matrix. They are used to find the probabilities with which each player should choose their strategies to maximize their expected payoff.

II. Probability -

9	7
5	11

$P_1 = \frac{6}{2+6} = \frac{6}{8} = \frac{3}{4}$ and $P_2 = \frac{2}{4+6} = \frac{2}{8} = \frac{1}{4}$

$Q_1 = \frac{4}{4+4} = \frac{4}{8} = \frac{1}{2}$ and $Q_2 = \frac{4}{4+4} = \frac{4}{8} = \frac{1}{2}$

III. Value f the game - Take 1st column value and the oddments of the 1st and 2nd row

$$V = \frac{(9*6)+(5*2)}{6+2} = \frac{54+10}{8} = \frac{64}{8} = 8$$

OR $$V = \frac{(7*6)+(11*2)}{6+2} = \frac{42+22}{8} = \frac{64}{8} = 8$$

The final conclusion is , the probability of Player A is $P_1 = \frac{3}{4}$ and $P_2 = \frac{1}{4}$

and Player B is $Q_1 = \frac{1}{2}$ and $Q_2 = \frac{1}{2}$

And the value of game is 8

Hence , the strategies of Player A $\left(\frac{3}{4}, \frac{1}{4}\right)$

Player B $\left(\frac{1}{2}, \frac{1}{2}\right)$

V = 8

<u>NOTE :-</u> Although each strategy has a lower than one chance of being chosen, their combined odds will be one.

2.2.1 DOMINANCE PROPERTY IN MIXED STRATEGY

By removing rows (or columns) that are dominated by other rows (or columns), the reward matrix size in various games can be decreased.

D.P for rows : $X \leq Y$

D.P for columns : $X \geq Y$

Problem :

Player B

		1	2	3	4
Player A	1	5	-3	3	4
	2	-4	5	4	5
	3	4	-4	-3	3

Row Min -3 ⟶ Maximin
-4
-4

Column 5 5 4 5
↓
Minimax

Maximin $\neq$ Minimax
i.e.; $-3 \neq 4$

To eliminate row , we should add the rows

$5 + (-3)+3+4 = 9$
$-4 + 5 +4 +5 = 10$
$4+(-4)+(-3)+3 = 0$

3rd row is dominated by 1st row as

$5> 4,\ -3> -4,\ 3> -3$ and $4 >3$

3rd row is eliminated

Player B

		1	2	3	4
Player A	1	5	-3	3	4
	2	-4	5	4	5
		1	2	7	9 ⟶ Max

Mixed strategies are two types

❖ Symmetric Mixed strategies

Every player employs the identical probability distribution across all strategies. To put it another way, players use the same set of techniques and randomize over them in the same manner. Players in these often used games have the same payoffs and strategies.

example: The Matching Pennies game is a well-known illustration of a game that employs symmetric mixed tactics.

Two players each have a penny in this game, and they can choose to reveal the heads or tails side. Whether or not the two pennies match determines the payout. Since each player selects their options at random with equal odds, the chances of matching are 1⁄2.

- **Asymmetric Mixed strategies**
- Players in this game employ a variety of probability distributions in their tactics. This is frequently employed in games where players have varying payoffs and strategies.

  For instance, think of the game of Rock, Paper, Scissors.
  Each player has the option to play rock, paper, scissors in this game. Every option ties with itself, wins against another option, and loses against another option. There is no Nash equilibrium in the pure strategies used by both players.
  There is a mixed strategy Nash equilibrium, nevertheless, if every player selects every option at random with an equal chance.

Many various kinds of games, including the well-known Prisoner's Dilemma and Battle of the Sexes, involve mixed strategies. They offer helpful resources for examining games with multiple equilibrium or in which players lack complete information.

2.2.2 GAMES WITH MIXED STRATEGIES

* No saddle point

1. Algebraic method
2. Graphical method
3. Linear programming method

	B	
	1	3
A 3	8	11
	8	11

	B
	1
A 3	8

Values of the game is 8
Player A adopt : 3 strategy
Player B adopt : 1 strategy

III. GAME WITHOUT SADDLE POINT

- Algebraic method or probability method

  › No saddle point
  › 2 * 2 matrix

› Principle of matrix can be used to reduce to 2 * 2

		Player B		
		B_1	B_2	Probabilities
Player A	A_1	a	b	p
	A_2	c	d	1 - p
Probability		q	1 - q	

Probabilities of A = p and 1 - p

$$p = \frac{a-c}{(a+d)-(b+c)}$$

Probabilities of B = q and 1 - q

$$q = \frac{a-b}{(a+d)-(b+c)}$$

Expected value of the Game V = $\frac{(ad-bc)}{(a+d)-(b+c)}$

Q. Solve the following game by probability method

B

		B_1	B_2	
A	A_1	3	5	3
	A_2	4	1	1
		4	5	

Max min = Mini max

$3 \neq 4$

So no saddle point

B

		B_1	B_2
A	A_1	3	5
	A_2	4	1

Player A

$$p = \frac{1-4}{(3+1)-(5+4)}$$

$$= \frac{-3}{-5} = \frac{3}{5}$$

$$1 - p = \frac{2}{5}$$

Player B

$$q = \frac{1-5}{-5}$$

$$= \frac{-4}{-5} = \frac{4}{5}$$

$$1 - q = \frac{1}{5}$$

$$V = \frac{ad-bc}{(a+d)-(b+c)}$$

$$= \frac{3-20}{-5} = \frac{17}{5}$$

- Graphical method

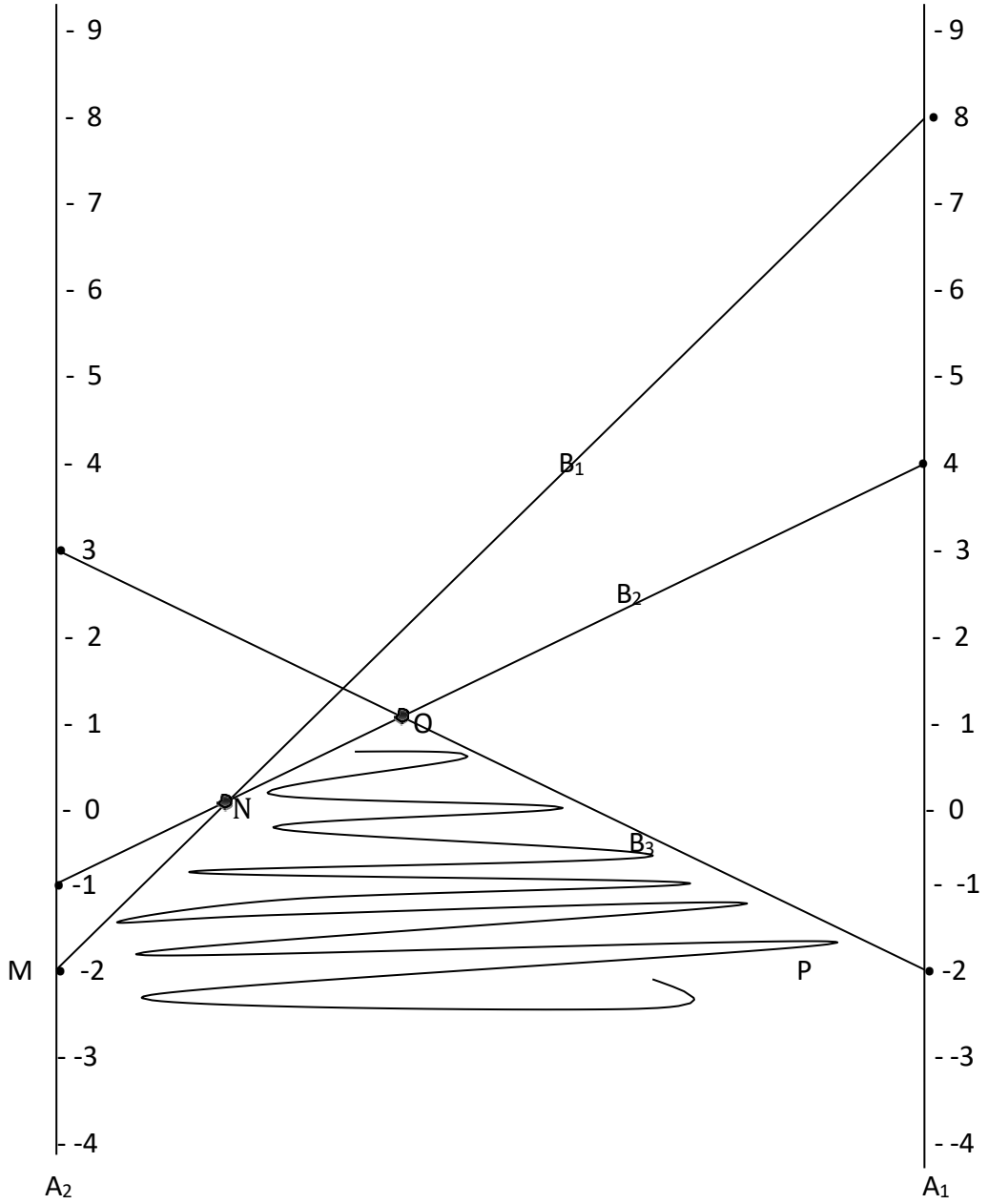
I. 2xn game- 2 row and n columns

II. mx2 game- m row and 2 columns

i.

Player B

		B_1	B_2	B_3
Player A	A_1	3	4	-2
	A_2	- 2	-1	3



O is the highest point

B_2 and B_3 are intersection point of O

So B_1 can ignore

		B	
		B_2	B_3
A	A_1	4	-2
	A_2	-1	3

And solve using probability method

- Player A :

$$p = \frac{3-(-1)}{(4+3)-(-1-2)}$$
$$= \frac{3+1}{7+3}$$
$$= \frac{4}{10}$$
$$= \frac{2}{5}$$
$$1 - p = 1 - \frac{2}{5}$$
$$= \frac{3}{5}$$

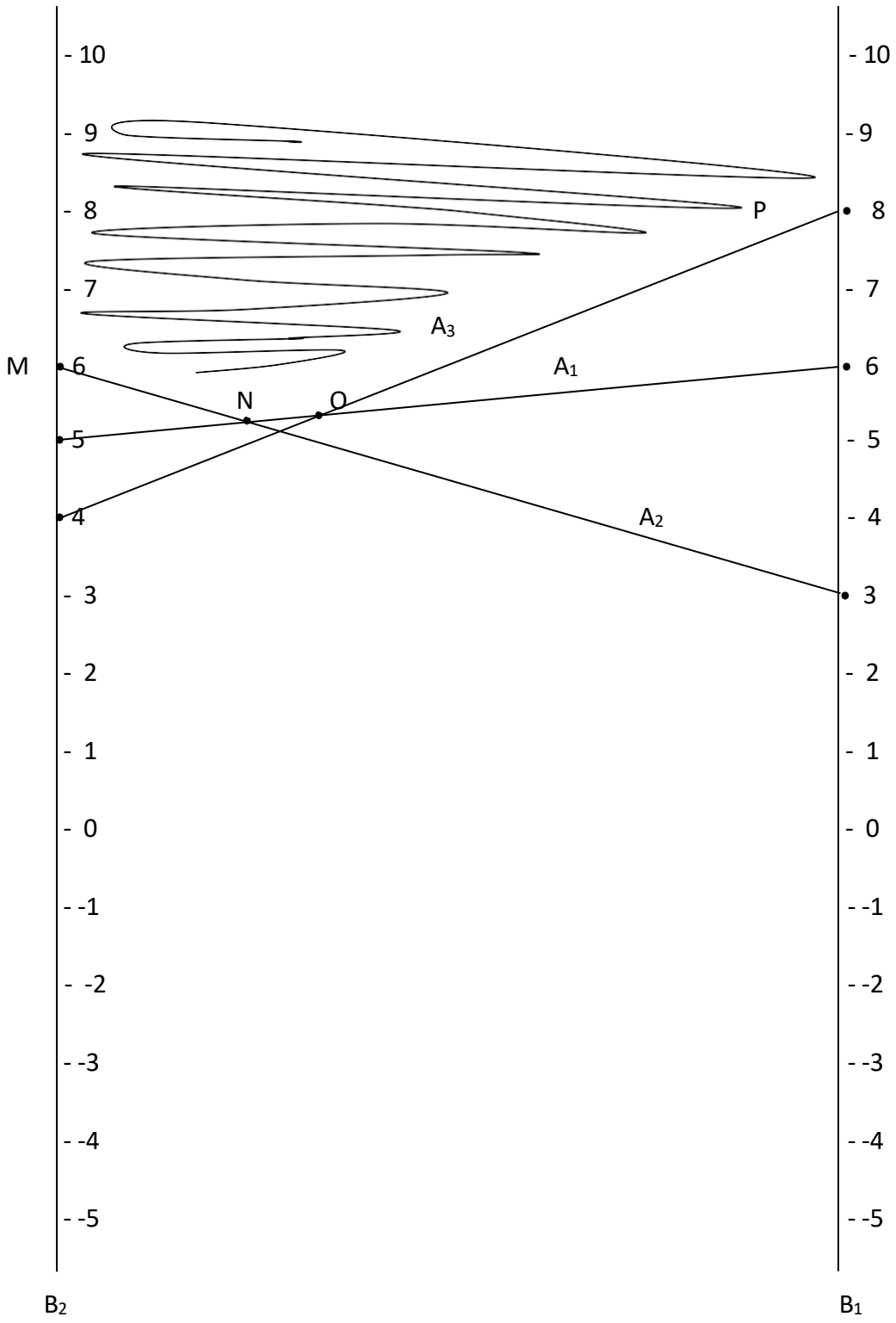
- Player B :

$$q = \frac{3-(-2)}{(4+3)-(-1-2)}$$
$$= \frac{3+2}{7+3}$$
$$= \frac{5}{10}$$
$$= \frac{1}{2}$$
$$1 - q = 1 - \frac{1}{2}$$
$$= \frac{1}{2}$$

And $V = \frac{12-2}{7+3} = \frac{10}{10} = 1$

ii.

		B	
		B_1	B_2
A	A_1	6	5
	A_2	3	6
	A_3	8	4



N is the lowest point

A_1 and A_2 are intersection point of N

So A_2 can ignore

		B	
		B_1	B_2
A	A_1	6	5
	A_1	3	6

And solve using probability method

- Player A :

$$p = \frac{6-3}{(6+6)-(5+3)}$$
$$= \frac{3}{12-8}$$
$$= \frac{3}{4}$$
$$1 - p = 1 - \frac{3}{4}$$
$$= \frac{1}{4}$$

- Player B :

$$q = \frac{6-5}{(6+6)-(5+3)}$$
$$= \frac{1}{12-8}$$
$$= \frac{1}{4}$$
$$1 - q = 1 - \frac{1}{4}$$
$$= \frac{3}{4}$$

And $V = \frac{36-15}{12-18}$

$$= \frac{21}{4}$$

2.3. PRINCIPLE OF DOMINANCE

The game's payoff matrix is made smaller by applying the principle of dominance, whereby a player's strategy can be eliminated if it is dominated by another strategy in all conditions.

I. Dominance property for rows:
   - If all the entries in a row should be less than or equal to the corresponding entries of another row, if yes, row can be deleted.
   - X and Y are row, if X less than or equal to Y then we can delete X.

II. Dominance property for columns:

- If all the entries in column should be greater than or equal to the corresponding entries of another column, if yes, column can be deleted
- X and Y are column, if X greater than or equal to Y then we can delete X.

Examples:(1)

PLAYER(B)

PLAYER(A)	X1	X2	X3	X4	X5	
Y1	4	6	5	10	6	31
Y2	7	8	5	9	10	39
Y3	8	9	11	10	9	47
Y4	6	4	10	6	4	30

- We can delete 4th row because 4th is less than 3rd row.
- Also we can delete 1st row because 1st row is less than or equal to 3rd row

PLAYER(B)

PLAYER(A)	X1	X2	X3	X4	X5	
Y2	7	8	5	9	10	
Y3	8	9	11	10	9	
	15	17	16	19	19	

- Now we can delete 5th column because 5th column is greater than the 1st ,2nd and 3rd columns.
- And we can also delete 4th column because 4th column is greater than the 1st and 2nd columns.
- And delete 2nd column because 2nd column is greater than 1st column.

PLAYER(B)

		X1	X3	
PLAYER(A)	Y2	7	5	12
	Y3	8	11	19

- Here we can delete 2nd row, because 2nd row is less than 3rd row

PLAYER(B)

		X1	X3	
PLAYER(A)	Y3	8	11	

- Then we can delete 3rd column, because 3rd column is greater than 1st column.

PLAYER(B)

		X1	
PLAYER(A)	Y3	8	

- Value of game is 8, player A adopt 3rd strategy and player B adopt 1st strategy.

CHAPTER 3

TYPES OF GAME

Games are categorized into different types. Some of them are as follows.

3.1. ZERO SUM GAME

When the game's net benefit is zero, it's called a zero sum game. The benefit of one individual is equal to the loss of another.

The sum of gain of one person (or group) and loss of another person (or group) is zero. Zero sum game are mostly solved with minimax theorem or with Nash equilibrium.

For example, football, where the one team will win and other will loss the net benefit of total game is zero.

Another example, consider two players, player(A) and player(B).

PLAYER(B)

		Y1	Y2
PLAYER(A)	X1	1	5
	X2	2	-1

Here X1, X2, Y1, and Y2 are the strategies of two players. And since it is a zero sum game,

- when player(A) choose X1 and player(B) choose Y1, then player(A) get an output 1 and player(B) get -1, that implies the net outcome is zero.
- when player(A) choose X1 and player(B) choose Y2, then player(A) get an output of 5 and player(B) get an output of -5, that implies the net outcome is zero.
- when player(A) choose X2 and player(B) choose Y1, then player(A) get an output of 2 and player(B) get an output of -2, that implies the net outcome is zero.
- when player(A) choose X2 and player(B) choose Y2, then player(A) get an output of -1 and player(B) get an output of 1, that implies the net outcome is zero.

3.2. NON-ZERO SUM GAME

A circumstance in which the game's net profit need not be zero is known as a non-zero sum game. Here one person's gain doesn't necessarily equivalent to the loss of other players. It's a win-win scenario. It has competitive and cooperative elements. We can see non-zero sum game in real life situations.

For example, a large group of people in an area is vaccinated against a disease, and their immunity can boost through this and reducing the chance of spreading of the disease and protecting even those who are not vaccinated. From this both individuals and society has gain, by reduced health care costs and better public health outcomes.

Most famous and classical example for non-zero sum game is prisoner's dilemma and battle of the sexes.

3.3. CONSTANT-SUM GAME

Constant-sum game is a game in which for any choice of strategies, the gain of one player and the loss of another player add up to a constant value c.

Consider an example,

where the constant sum is 10

PLAYER(B)

		X1	X2
PLAYER(A)	Y1	8	6
	Y2	3	7

- When player(A) choose Y1 and player(B) choose X1, player(A) gets an output of 8 and player(B) gets an output of 2, that implies the net outcome (8+2) is 10.
- When player(A) choose Y1 and player(B) choose X2, player(A) get an output of 6 and player(B) get an output of 4, that implies the net outcome (6+4) is 10
- When player(A) choose Y2 and player(B) choose X1, player(A) get an output of 3 and player(B) get an output of 7, that implies the net outcome (3+7) is 10
- When player(A) choose Y2 and player(B) choose X2, player(A) get an output of 7 and player(B) get an output of 3, that implies the net outcome (7+3) is 10

3.4. COOPERATIVE GAME

Negotiations and agreements between players are used in cooperative or negotiation games to persuade players to pursue a specific approach.

- Players negotiate their contracts to that they can make joint strategies
- For example, the game played between the buyer and seller or, firms buying row material or services can negotiate for price settlements.
- In prisoner's dilemma both decided to remain silent by negotiation

3.5. NON-COOPERATIVE GAME

Non-cooperative game or non-negotiation game, it refers to game in which the players decide on their own strategy to maximize their gain.

- It is a game in which no form of negotiation and binding contracts
- Here result is accurate and it is a deep analysis.
- In order to optimize their earnings, players choose their own tactics.
- For example, while setting up a price of a product.
- Prisoner's dilemma is an example for this type game also. Here no negotiations are happened in between the prisoners.

3.6. N-PERSON

An n-person is a strategic-scenario where players are involved, each decision that affect both their own outcomes and outcome of others, or we can simply say that n-person game is which involved n persons in a game.

A classical example, for n-person game is the prisoner's dilemma extended to multiple players. We can see n-person by other example.

Let us take 3 players and each have two strategies. The 3 players are friends and they discussing how to spend their free time.

Player (1) prefer to go a football game and player (2) prefer to movie. Let player (3) matches with anyone either movie or football,

		Player (1)	
		football	movie
Player (2)	football	(3,1,2)	(0,2,1)
	movie	(2,0,1)	(1,3,2)

- If player (1) and player (2) choose to go football then player (1) will get 3 and player (2) get 1, player (2) get 2 since player (3) choose player 1 choice.
- If player (1) prefer football but choose to go movie since player (2) prefer it player (1) get 0 and player (2) get 2, player (3) get 1.
- If player (1) go to movie again prefer so, get 2 and player (2) get 0 since they prefer to movie but go to football and player 3 will get 1 since player (3) match with player.
- If player (1) and player (2) go to movie, player (1) get 1 and player (2) get 3 and player (3) get 2. Since player (3) match player (2).

Each player payout is indicated in a separate cell. Player 1 payoff is represented by the first number, Player 2 payoff by the second, and Player 3 payoff by the third.

From matrix there are two Nash equilibrium.

1. Player 1 and player 2 choose football and player 3 also choose football. This case is stable because no player has an initiative to deviate.
2. Player 1 choose movie and player 2 choose movie so player 3 also choose movie. This is also stable.

The conclusion is player 3 will likely to choose (F,F,F) or (M,M,M) because these strategies. Represent Nash equilibrium of game, where no player can improve their outcome by changing their own choice unilaterally.

3.7. 2X2 TWO-PERSON GAME

A strategic game between two players where each player has 2 strategies. The outcome is represent using 2x2 matrix and showing the payoff for both players depending on their strategy choices. Each cells in the matrix contain a payoff pair. The game can be zero-sum or non-zero sum.

Example:

Let us take two players, player A and player B. there are two strategies for each player where fight(F) and avoid(A),

	F	A
F	(1,1)	(3,0)
A	(0,3)	(2,2)

- If both players choose fighting gives low preference.
- If one avoids while the other fights, the fights get more and not stable.
- If player A and player B is avoid fights. Then the solution is stable.
- Neither the player can switch to get a better outcome, so (A, A) = (2, 2) is the Nash equilibrium.

3.8. mX2 GAME

mx2 game is opposite of 2xm game, where player 1 has m strategies and player 2 has two strategies. It is similar to 2xm game but with role reversed.

Example:

Let us take two players Alice and Bob. Alice has 3 restaurants choices like Italian, Mexican and Chinese. Bob still choose between Italian and Mexican. They prefer to eating together. If they go different place they get 0,

		Bob Italian	Bob Mexican
Alice	Italian	(3,2)	(0,0)
Alice	Mexican	(0,0)	(2,3)
Alice	Chinese	(0,0)	(1,1)

There are two pure strategy Nash equilibrium in 3x2.

- (Italian, Mexican)
- (Mexican, Mexican)

(Chinese, Mexican) is not equilibrium, because Alice would switch to get a better payoff.

3.9. 2 - PERSON GAME

If the total payouts to each player in a two-person game are the same for every possible outcome, the game is said to be zero-sum.
Each of the two players has a unique set of strategies. The goal of one player is to maximize her payout, while the other player tries to decrease it. In actuality, a player's gain is another's loss.

Key concepts include the payoff matrix, which shows payoffs for each possible combination of actions, and No actor may unilaterally alter their tactics to increase their payoff in a Nash equilibrium.

One player gives the other a payout as part of their competition. Because the reward to both players for any given scenario comes up to zero, and because one player's loss is the other player's gain, these games are known as zero-sum games.

For 2 person game ,the prisoner's dilemma a classical example . We can see 2 person by other example

Example:

Coin Matching Colleen plays a game with Roger. Each of them has a coin. They will both exhibit opposing points of view at the same time. If both display heads, there will be no money exchanged. If Roger shows heads and Colleen shows tails, Colleen will give him $1. If Colleen

shows heads and Roger shows tails, Roger will give her $1. If both show tails, Colleen will give Roger $2.

The two players in this two-player game are Roger and Colleen. It's a zero-sum game as well. This suggests that when Roger wins, Colleen loses.

Colleen

		Heads	Tails
Roger	Heads	Roger pays/gets $0 Colleen pays/gets $0	Roger gets $1 Colleen pays $1
	Tails	Roger pays $1 Colleen gets $1	Roger gets $2 Colleen pays $2

Because each player's winnings are equal to the opposite of the opponent's winnings in any given scenario, this game is known as a two-person, zero-sum game. The pay-off for that player is the sum of money won by either player in any certain circumstance.
Only the payoff for the row player in each scenario is displayed in the game's pay-off matrix.

Colleen

R		H	T
o	H	0	1
g	T	-1	2

A strategy is a player's plan of action against their opponent.
There are two different strategies for each player: H and T. Assuming both participants wish to maximize their payout, we will attempt to identify each player's optimal strategy.

Each player in the current scenario has two options for strategies: H and T. For instance, if Roger uses strategy H and Colleen uses strategy T, the result is 1: Roger wins $1 and Colleen loses $1. Assuming that both players wish to maximize their payoff, our objective is to identify the optimal approach for each player.

When we look at gamers who are playing the same game over and over again, our conclusions will make the most sense.

CHAPTER 4

APPLICATIONS OF GAME THEORY

4.1. Application Of Game Theory in Economics

Game theory was originally introduced to solve various problems in economics. The theory of games was formulated by economists Neumann & Morgenstern, who were concerned with policymaking decisions to save the country from the adverse effects of trade cycles and depression and to ensure the smooth way of development. Game theory is an important device that shows the way how to come out of the crises and depression that engulfed the economy.

One crucial strategy used in mathematical economics is game theory. Game theory is a method used by economists to study economic competition and phenomena including voting theory, experimental economics, political economy, behavioural economics, mechanism design, bargaining, and auctions.

By resolving significant issues with earlier mathematical economic models, game theory revolutionized the field of economics. Game theory is frequently used by economists to describe the behavior of oligopoly firms. Predicting probable results when businesses participate in specific behaviors, such price fixing and collusion, is helpful.

4.2. Application Of Game Theory in Politics

In essence, game theory is a method for comprehending group decision-making. One of the most well-known instances of game theory is Prisoner's Dilemma, which looks like a normal law and order situation.

Game theory as offers valuable insights into the complex strategic decision-making process that governs interactions between nations, international organizations, and non-state actors. It also allows policymakers to interpret various factors that influence the actions and choices of leaders on the international stage. The relevance of game theory is more pronounced in international politics than in domestic politics.

Here are some instances of game theory in politics:

- The strategies of most countries participated in the world war could be expressed in terms of games.

Every nation was busy finding strategies to cope with the move taken by its ene my countries.

- Nations make alliances or coalition, like a military alliance whose major objective is to gain control over international politics.

4.3. Application Of Game Theory in Business

Despite being theoretical in origin, game theory is a mathematical instrument that has a significant impact on real-world circumstances. Game theory may analyze the goals, tactics, and outcomes of each side in business choices, which frequently entail cooperation and competition between several parties.

Data on corporate rivalry is gathered, arranged into a benefit matrix, and compared in order to apply game theory to commercial decision-making. According to game theory's prisoner's dilemma model, win-win cooperation is the best course of action when it comes to pricing because a prolonged price war will ultimately result in losses for both sides.

By working on the prisoner’s dilemma, we could understand that cooperation is the best course of action when it comes to pricing since it can offer a successful and efficient strategy for all businesses to profit from each other's growth. Additionally, entrepreneurs can use game theory to make wise business decisions on future enterprise development and reform as well as corporate collaboration and competitiveness.

4.4. Application Of Game Theory in Biology

Many facets of human behaviour have been explained by game theory. In biology, game theory aims to forecast social behaviour and other characteristics that affect how people interact.

Numerous issues in botany and zoology can be solved using game theory. Animal fighting, cooperation, communication, coexistence of alternative features, mating systems, sex conflict, offspring sex ratio, and individual distribution within their habitat are all scenarios that are interpreted using game theory in zoology. Game theory is applied in botany to analyze sex allocation, power size, root competition, nectar production, seed germination, and dispersal..

4.5. Application Of Game Theory in Project Management

An organization's time-bound exercise to provide a good service or outcome is called a project. A sensible decision-making process is essential for the success of the project.

Game theory can be a useful tool in project management, particularly in dealing with risk management, stakeholder engagement, and complex decision-making. Project managers can better grasp possible conflicts, opportunities, and cooperative processes, as well as the best ways to make decisions in the face of ambiguity, by using game theory to observe the direct interaction between different stakeholders.

Through the prediction of stakeholder reaction, game theory assists project managers in making more practical decisions. Additionally, by utilizing game theory, they can identify more effective solutions; they can find more effective solutions by modelling conflicts as geneses and taking into account the point of view of each stakeholder.

Using game theory is also helpful for the project manager to allocate resources effectively to maximize the output and cooperation of team members. Game theory also shows the importance of monitoring and adaptation in dynamic environments. By applying game theory, project managers can understand the potential risks by analysing the interaction between stakeholders and their strategic moves throughout the project cycle

4.6. Application Of Game Theory in Computer Science

Game theory has a wide range of application computer science which includes artificial intelligence, machine learning, network security and distributed systems. The main benefits of it are improved decision making, increased security and more efficient resource allocation.

In the case of game playing AI strategic decision making, imperfect information game, game tree search AI opponent modeling is some of the application. The benefit of it includes improved games playing increased challenge and better AI decisions making. Example of game-playing AI are Alpha Go and Poker AI

In machine learning, Game theory can be used to create adversarial competition to build robust models. On the other hand, participants defend against adversarial attacks also. Also, it can help to improve overall security and robustness in red world applications. Examples of machine learning are: adversarial learning, which include a Generative Adversarial networks and adversarial training and multi-Agent learning, that includes multi agent reinforcement learning and game theoretic models.

In network security it includes intrusion, detection and security protocols which game theory helps to detect, predict and protest against cyber-attacks.

Example of network security

- Intrusion: detection which includes Game theoretic models and optimal defense strategies
- Security protocol design includes game theoretic security protocols and cryptographic protocols.

- Distributed system includes resources allocation and machine design it also helps to improve security and better decision making where each component have their own goals and may not always cooperate.

Examples of distributed systems

- Resources allocation which includes cloud computing and distributed resource allocation.

- Network formation which includes network topology formation and peer-to-peer network etc.

4.7. Application Of Game Theory in Real Life

As we know that the concept of game theory has a great relevance not only to mathematicians but also physicists and theorists. Just like this, the principle of the theory is also embedded in our daily lives too.

In real life some of the important application of game theory include decision making strategic thinking, conflict resolution etc. where as in the case of social interaction, it includes social networks, bargaining cooperation, social interaction.

Everyday decisions: -

- Rush hour traffic
- Shopping
- Auctions

Social Interactions: -

- Dilemma
- Cooperation

Also in personal relationships, it is used to build trust, resolve conflict and improve communication.

Examples:

Personal relationships

- Friendships
- Negotiating with family members

In board games the benefit of application of game theory includes improved game play, strategic decision making, analytical skills help each player to identify the optimal outcomes.

Strategic board games include:

Chess, go and risk,

Economic board games include:

Monopoly and game of life,

Other board games include: poker, scrabble etc. are some of the examples.

CONCLUSION

Game theory is a branch of mathematics and economics, provide a framework for analyzing strategic interaction where outcome for each participant depend on the choice of all. In game theory strategies are planned set of actions that players choose to maximize their outcome or minimize losses. The strategies can be pure or mixed on the basis pf game.

Game theory is defined by several key features like multiple decision-makers, set of strategies and clear defined payoffs. Game can be zero sum or non-zero sum, cooperative or non-cooperative. Game theory assume rationality among players, meaning they act in their own best interest with information availability.

The application of game theory is vast. In economics it shows how to come out. From the crises and depression situation. Also game theory important in international politics. In botany game theory is used to analyze seed dispersal, seed germination etc. In real life game theory help to decision making, strategic thinking etc.

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